

ANTONIO BARBOSA

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SUMMARY

Senior Full-Stack Engineer and Technical Architect with over a decade of experience designing and building large-scale distributed systems, AI-driven features, and high-performance backend architectures. Specialized in RAG pipelines, embeddings, vector search, and LLM integration, combining deep backend expertise with practical AI engineering. Proven track record delivering complex systems end-to-end, from research and architecture to implementation and optimization, across SaaS platforms, real-time synchronization engines, and enterprise-grade applications. Skilled in TypeScript, Node.js, Ruby on Rails, React, and cloud infrastructure. Operates with high autonomy, strong ownership, and a focus on creating reusable building blocks that accelerate teams and scale across products.

EXPERIENCE

Full-Stack Engineer | Technical Architect

RIIPEN

March 2022 – November 2025, Vancouver, BC

- Architected and delivered the company's first full RAG and embedding pipeline, powering 6+ new AI features and improving content discovery accuracy by ~30%.
- Designed vector-similarity search infrastructure and built reusable service abstractions, reducing development time for new AI features by ~40%.
- Implemented large-scale data ingestion, cleaning, and embedding generation pipelines, increasing processing throughput by 3×.
- Rebuilt the Persona document-verification workflow, redesigning the state machine, callback model, and error-handling architecture, reducing onboarding failures.
- Implemented security-focused callback validation and retry logic, eliminating exploitation attempts related to callback manipulation.
- Collaborated with product and design to rewrite frontend flows in React, ensuring consistent state transitions and lowering validation support tickets by ~25%.
- Refactored legacy Rails modules to support AI-driven features, improving system reliability and reducing performance bottlenecks.
- Owned end-to-end delivery of backend and infrastructure improvements, improving API performance and stability for thousands of daily users.

Senior Software Developer

CGI

October 2020 – November 2021, Drummondville, QC

- Developed and maintained backend services in TypeScript and Node.js, improving system stability and reducing recurring errors by an estimated 15–20%.
- Integrated Node.js services with Java enterprise systems, supporting reliable data flow across 5+ internal applications used daily by multiple teams.
- Implemented API integrations and data-processing pipelines that automated a significant portion of manual workflows, reducing operational overhead by roughly one-third.
- Improved accessibility by adding keyboard navigation and ARIA support to about 40% of the platform, expanding usability for keyboard-only and assistive-technology users.
- Collaborated with distributed teams across three time zones, accelerating cross-system issue resolution and improving delivery speed for shared features.
- Optimized enterprise frameworks and service patterns, contributing to measurable improvements in response time and long-term maintainability.

Full-stack Developer & Solutions Architect | Technical Lead

INTERATIVA

September 2016 – January 2021, São Paulo, Brasil

- Led technical architecture and full product ownership across more than 100 projects, building reusable platform components, distributed systems, and long-term foundations reused across teams. Coordinated Unity, backend, and full-stack teams while delivering solutions for major clients such as Samsung, Pfizer, Merck, Volkswagen, and AstraZeneca.
- Researched and architected a complete offline-first distributed access-control system (VOLPO), designing the full end-to-end flow including RFID integration, event queues, WebSocket synchronization, conflict-resolution strategies, and real-time distributed state. Implemented the backend in Node.js with MongoDB and real-time sockets and delivered the frontend in Angular/React. The system became the company's flagship product, supporting large-scale events with thousands of attendees and reducing on-site operational workload by a significant margin. The synchronization engine and supporting modules powered multiple subsequent projects and became part of the company's core architecture.
- Migrated a legacy REST stack to a full GraphQL architecture, enabling more flexible data access and improving development speed across 100+ projects. Designed scalable schemas, efficient resolvers, reusable patterns, declarative data models, and real-time subscriptions, contributing to smoother data flows and noticeably faster iteration for cross-team development.
- Coordinated Unity development efforts and built integration layers between Unity (C#) and backend servers, enabling real-time interactions for multiple interactive experiences. Delivered mini-games and dynamic applications used in corporate events, producing reusable UI component systems that reduced UI development time by an estimated 30–40% across teams.
- Developed foundational building blocks used throughout the organization, including the central synchronization engine with retry logic, payload versioning, reconnection handling, and offline merge algorithms. Created reusable UI systems for real-time data visualization, accelerating interface delivery across 240+ active repositories and eliminating repetitive rebuild cycles.

- Contributed to a large and active codebase with over 240 repositories and delivered solutions that supported dozens of large-scale corporate events and applications.

PROJECT

GAME JAMS

Personal Project • January 2018 – December 2019

- Led the technical direction for **two complete game prototypes**, ensuring each delivered a continuous **2-hour gameplay loop** within the 48–72 hour jam window.
- Designed and implemented core gameplay mechanics and interaction systems that kept players engaged through **90% of the implemented checkpoints**, with some users fully exploring **100%** of the available content.
- Organized team workflow and unblocked designers and developers, reducing bottlenecks and helping maintain steady progress throughout the jam cycle.
- Built rapid-iteration features in Unity (C#), enabling **multiple gameplay updates per day** and faster validation of mechanics, pacing, and user flow.

PUMP IT UP TOURNAMENT PLATFORM

(Personal Project) • November 2018 – December 2018

- Automated player randomization and tournament bracket generation, replacing a fully manual workflow typically requiring **30–60 minutes** per event.
- Built a music randomizer that increased match variety and eliminated manual selection errors throughout tournaments.
- Implemented real-time notifications showing who currently playing → **showing active players in real time**, improving event flow and reducing coordination overhead by **40–50%**.
- Designed a centralized interface that streamlined registration, scoring, and match tracking, accelerating event operations and improving consistency across matches.

EDUCATION

Post-Graduate Certificate in Web Development

Minor in Did you minor in anything? • Fatea • Lorena SP • 2014

Technology Degree in Systems Analysis and Development

Fatec • Guaratinguetá, SP (Brazil) • 2012–2014

- Developed my college's project-submission system, enabling **100% of projects** to be submitted online.
- Built user flows, submission logic, review tools, and administrative dashboards to streamline the entire process.
- Reduced manual review work for professors and improved overall organization and accessibility of submissions.

3 years of Chemical Engineering

USP • Lorena SP (Brazil) • 2008

- Primarily self-taught through hands-on projects

SKILLS

Node.js, Express, Ruby on Rails, .NET Framework, REST API design, GraphQL

RAG, Embeddings and vector similarity, LLMs, MCP, Embedding generation and consumption pipelines, Vector database integration, Semantic search

TypeScript, JavaScript, HTML5, CSS/SCSS, Tailwind CSS, Bootstrap

React, Angular, Vue, Ember.js, REST API, GraphQL, WebSockets

PostgreSQL, MongoDB, MySQL, SQL Server, Vector databases, Oracle SQL

Python, C#, PHP, Java, Unity, C# (Unity scripting)

AWS, Heroku, Docker, CI/CD, Git, GitHub, Nginx configuration, Server architecture, Automation and tooling

Raspberry Pi, RFID, Arduino, Hardware-software integration, Custom protocols for hardware communication

RSpec, Jest, JUnit, Integration testing, TDD, Test automation

Distributed systems architecture, Microservices design, Offline-first systems, Real-time data synchronization, Event-driven architecture, State machine design, System scalability and performance optimization, Security best practices, Design Patterns, SOLID principles, Clean Code, OOP and functional programming paradigms, Jira, Kanban, Scrum, Agile methodologies, Code review, Technical mentorship, Technical documentation

Portuguese, English, French
